

HMMT 2013
Saturday 16 February 2013
HMIC

1. [5] Let $\mathcal{S}$ be a set of size n , and k be a positive integer. For each $1 \leq i \leq kn$, there is a subset $S_i \subset \mathcal{S}$ such that $|S_i| = 2$. Furthermore, for each $e \in \mathcal{S}$, there are exactly $2k$ values of i such that $e \in S_i$. Show that it is possible to choose one element from S_i for each $1 \leq i \leq kn$ such that every element of $\mathcal{S}$ is chosen exactly k times.

Answer: N/A Consider the undirected graph $G = (\mathcal{S}, E)$ where the elements of $\mathcal{S}$ are the vertices, and for each $1 \leq i \leq kn$, there is an edge between the two elements of S_i . (Note that there might be multiedges if two subsets are the same, but there are no self-loops.) Consider any connected component C of G , which must be a $2k$ -regular graph, and because $2k$ is even, C has an Eulerian circuit. Pick an orientation of the circuit, and hence a direction for each edge in C . Then, for each i such that the edge corresponding to S_i is in C , pick the element that is pointed to by that edge. Since the circuit goes into each vertex of C k times, each element in C is picked exactly k times as desired. Repeating for each connected component finishes the problem.

2. [7] Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for all real numbers x, y ,

$$(x - y)(f(x) - f(y)) = f(x - f(y))f(f(x) - y).$$

Answer: $f(x) = 0, f(x) = x$ First, suppose that $f(0) \neq 0$. Then, by letting $x = y = 0$, we get that either $f(f(0))$ or $f(-f(0))$ is zero. The former gives a contradiction by plugging in $x = f(0), y = 0$ into the original equation. Thus, $f(-f(0)) = 0$.

Now plugging in $x = 0, y = -f(0)$ gives that $f(2f(0)) = f(0)$. Finally, we note that by symmetry of the left hand side of the functional equation, $f(x - f(y))f(f(x) - y) = f(y - f(x))f(f(y) - x)$, and letting $y = 0$ and $x = 2f(0)$ we get $f(f(0))f(f(0)) = f(-f(0))f(-f(0)) = 0$, giving and so $f(f(0)) = 0$, again a contradiction. Therefore, $f(0) = 0$.

Setting $y = 0$ yields $xf(x) = f(x)f(f(x))$, so if $f(x) \neq 0$, then $f(f(x)) = x$. On the other hand, setting $y = f(x)$ yields $(x - f(x))(f(x) - f(f(x))) = f(x - f(f(x)))f(0)$, so in fact $f(f(x)) = f(x) \rightarrow f(x) = x$. Thus, for all x , $f(x) = 0$ or $f(x) = x$.

Clearly, $f \equiv 0$ satisfies the original equation, so suppose there exists a non-zero x with $f(x) = x$. Then, for all $y \neq x$, if $f(y) \neq 0$, the original equation yields $x(x - y) = xf(x - y)$. Thus, $f(x - y) = x - y$. Similarly, if $f(y) = y$, we find $(x - y)^2 = f(x - y)^2$, so $f(x - y) = x - y$. It follows that f must be the identity, which indeed is a solution to the original equation. The proof is complete.

3. [8] Triangle ABC is inscribed in a circle ω such that $\angle A = 60^\circ$ and $\angle B = 75^\circ$. Let the bisector of angle A meet BC and ω at E and D , respectively. Let the reflections of A across D and C be D' and C' , respectively. If the tangent to ω at A meets line BC at P , and the circumcircle of APD' meets line AC at $F \neq A$, prove that the circumcircle of $C'FE$ is tangent to BC at E .

Answer: N/A We will show that $CE^2 = (CF)(CC')$. By a simple computation using the given angles, one may find that this is equivalent to $CF = AC - AB$, or $AF = 2AC - AB$.

We compute AF by trigonometry. Assume for simplicity that $AC = \frac{1}{2}$, so $AD' = 2AD = 2AC = 1$ because $\triangle ACD$ is isosceles by angle chasing. We first compute PD' by the law of cosines on triangle APD' , which yields $PD'^2 = AP^2 + AD'^2 - 2(AP)(AD')\cos 75 = AP^2 + 1 - 2AP\cos 75$. We may easily compute AP by the law of sines in triangle BAP to be $\frac{1}{\sqrt{2}}$. Thus, $PD'^2 = \frac{1}{2} + 1 - \sqrt{2}\frac{\sqrt{6}-\sqrt{2}}{4} = \frac{4-\sqrt{3}}{2}$.

By the law of sines within the circumcircle of $APD'F$, we have $\frac{FD'}{\sin 30} = \frac{PD'}{\sin 75}$. From this, we find that $FD' = \frac{PD'\sin 30}{\sin 75} = \frac{4-\sqrt{3}}{\sqrt{2}+\sqrt{6}}$. Thus, we have by the law of cosines on triangle $D'AF$ that $AF^2 + AD'^2 - 2(AF)(AD')\cos 30 = AF^2 + 1 - \sqrt{3}AF = FD'^2 = \frac{4-\sqrt{3}}{4+2\sqrt{3}}$. Finally, solving for AF yields $AF = \frac{\sqrt{3} \pm (2\sqrt{3}-3)}{2}$ from which we indeed get $AF = \frac{3-\sqrt{3}}{2}$ which equals $2AC - AB$.

4. [10] A subset $U \subset \mathbb{R}$ is *open* if for any $x \in U$, there exist real numbers a, b such that $x \in (a, b) \subset U$. Suppose $S \subset \mathbb{R}$ has the property that any open set intersecting $(0, 1)$ also intersects S . Let T be a countable collection of open sets containing S . Prove that the intersection of all of the sets of T is not a countable subset of $\mathbb{R}$.

(A set Γ is *countable* if there exists a bijective function $f : \Gamma \rightarrow \mathbb{Z}$.)

Answer: N/A If S is uncountable then we're done, so assume S is countable. We may also assume that the supersets are a chain $V_1 \supset V_2 \supset V_3 \supset \dots$ by taking intersections.

We will use the following fact from point set topology:

If $K_1 \supset K_2 \supset \dots$ is a sequence of nonempty compact sets, then their intersection is nonempty.

Now, we construct an uncountable family of such sequences $K_1 \supset K_2 \supset \dots$ such that K_i is a nontrivial (i.e. $[a, b]$ with $b > a$) closed interval contained in V_i , and any two such sequences have disjoint intersection. The construction proceeds as follows. At each step, we choose out of countably many options one closed interval $K_{i+1} \subset K_i$.

To choose K_1 , we claim that there exist countably many disjoint closed intervals in V_1 . To do this, choose an open interval in V_1 of length at most $1/2$, and take some closed interval inside it. In the remainder of $(0, 1)$, which has measure at least $1/2$ and where V_1 is still dense, choose another open interval of length at most $1/4$, and a closed interval inside it. This process can be repeated indefinitely to find a countably infinite family of nontrivial disjoint closed intervals in V_1 . Choose one of them to be K_1 .

Now, inductively apply this construction on K_i to find countably many choices for K_{i+1} , all disjoint. As a result, we get a total of $\mathbb{N}^{\mathbb{N}}$, i.e. uncountably many, choices of sequences K_i satisfying the given properties. Any pair of sequences eventually have disjoint K_i after some point, so no two intersections will intersect. Thus we can choose a point out of the intersection of each K_i sequence to find uncountably many points in the intersection of all the V_i , as desired.

5. [12]

- (a) Given a finite set X of points in the plane, let $f_X(n)$ be the largest possible area of a polygon with at most n vertices, all of which are points of X . Prove that if m, n are integers with $m \geq n > 2$, then $f_X(m) + f_X(n) \geq f_X(m+1) + f_X(n-1)$.
- (b) Let P_0 be a 1-by-2 rectangle (including its interior), and inductively define the polygon P_i to be the result of folding P_{i-1} over some line that cuts P_{i-1} into two **connected** parts. The *diameter* of a polygon P_i is the maximum distance between two points of P_i . Determine the smallest possible diameter of P_{2013} .
- (In other words, given a polygon P_{i-1} , a fold of P_{i-1} consists of a line l dividing P_{i-1} into two connected parts A and B , and the folded polygon $P_i = A \cup B_l$, where B_l is the reflection of B over the line l .)

Answer: N/A

Note from <http://artofproblemsolving.com/community/c129h529051p3039219>. Minor slip: In Section 1.3, Case 2 (for #5), how are B' and A' defined exactly? Right now it's not clear to me that they have $m+1$ and $n-1$ vertices, respectively, since it's not always possible to find $B_i, B_{i'}$ with $f(i') - f(i) = i' - i$, is it? (The simplest counterexample is when $m = 3$, $n = 5$, and the A_j, B_k are pairwise distinct.)

[b]Edit:[/b] I think it should be $f(i') - f(i) = i' - i + 1$ (which is not hard to find using a discrete continuity argument) and then $B' = B_1 \dots B_i A_{f(i)+1} \dots A_{f(i')} B_{i'+1} \dots B_m$ (A' the complement of B'), in which case everything seems to work out. (We are assuming at most two points lie between B_k, B_{k+1} for all k ; otherwise we can just use Case 1.) Thanks to Tim for clarifying most of this! I probably should have tried the $m = 3$, $n = 5$ case a little harder before posting this, since it generalizes fairly easily.

1 Convexity of the largest area of a polygon function

(a)

Let V be a finite set of points in the plane. Let f_V be a function that takes integers ≥ 3 as input, and outputs a polygon of largest area with vertices in V .

Lemma 1.1. $[f_V(a+1)] + [f_V(a-1)] \leq 2[f_V(a)]$, where $[P]$ represents the area of polygon P .

Proof. Let the vertices of $f_V(a+1)$ be $1_{a+1}, 2_{a+1}, \dots, (a+1)_{a+1}$, and let the vertices of $f_V(a-1)$ be $1_{a-1}, 2_{a-1}, \dots, (a-1)_{a-1}$.

Lemma 1.2. *The collection of vertices*

$$W = (1_{a-1}, 2_{a-1}, \dots, (a-1)_{a-1}, 1_{a+1}, \dots, (a+1)_{a+1})$$

. It's clear that $W \subset V$. are in convex configuration.

Proof. Suppose that the vertex $s_x \in W$ (for some integers $x \in \{a-1, a+1\}, s$) is inside the convex hull of W . Then there exists a vertex k on the convex hull of W such that replacing vertex s_x in $f_V(x)$ with k increases the area of the polygon $f_V(x)$. However, $f_V(x)$ is defined to be the polygon with x sides with maximum area. Contradiction. This proves Lemma 2. $\square$

Therefore the vertices in W are in convex configuration. Label them with $1, 2, \dots, 2a$ in clockwise order, starting from an arbitrary vertex. Let $(2a)$ denote the vertex labeled $(2a)$. (The vertex (1) will be shorthanded as 1 when there is no ambiguity).

Lemma 1.3. *We have the inequality*

$$[f_V(a+1)] + [f_V(a-1)] \leq [\mathcal{P}1357 \dots (2a-1)] + [\mathcal{P}2468 \dots (2a)].$$

(Throughout, the symbol $\mathcal{P}$ denotes an arbitrary polygon.)

Proof. It suffices to show that

$$\begin{aligned} & 2[\mathcal{P}1234 \dots (2a)] - [f_V(a+1)] - [f_V(a-1)] \\ & \geq 2[\mathcal{P}1234 \dots (2a)] - [\mathcal{P}1357 \dots (2a-1)] - [\mathcal{P}2468 \dots (2a)]. \end{aligned}$$

We will do this using the next few lemmas.

Lemma 1.4. *Let x and y be integers $\leq 2a$. If ray $\overrightarrow{(x)(x+1)}$ intersects (or is parallel to) ray $\overrightarrow{(y)(y-1)}$. Then $[\mathcal{P}(x)(x+1)(x+2) \dots (y-1)(y)] \geq [\triangle(x)(x+1)(x+2)] + [\triangle(x+1)(x+2)(x+3)] + \dots + [\triangle((y-2)(y-1)(y))]$ (where $(2a+k)$ is equivalent to (k) for all k).*

Proof. $[\mathcal{P}(x)(x+1)(x+2) \dots (y-1)(y)] = [\triangle(x)(x+1)(x+2)] + [\triangle(x)(x+2)(x+3)] + \dots + [\triangle(x)(y-1)(y)]$. Suppose ray $\overrightarrow{(x)(x+1)}$ intersects ray $\overrightarrow{(y)(y-1)}$ at point D , the convexity of W tells us that all vertices in between x and y exclusive going clockwise from x are contained inside triangle $(x)D(y)$. It follows that for any x' between x and y exclusive going clockwise from x , that (x) is farther from line $\overrightarrow{(x')(x'+1)}$ than $(x'-1)$. Therefore

$$[\triangle(x)(x')(x'+1)] \leq [\triangle(x'-1)(x')(x'+1)].$$

Summing these relations over all x in between x and y exclusive going clockwise from x gives the statement of Lemma 4, as desired.

An identical argument works if ray $\overrightarrow{(x)(x+1)}$ and ray $\overrightarrow{(y)(y-1)}$ are parallel, except instead of D , all the points between x and y are bounded by the two rays and the line $\overrightarrow{(x)(y)}$.

This proves Lemma 4. $\square$

Let z and s be integers such that $z \in \{a-1, a+1\}$ and both vertices s_z and $(s+1)_z$ are adjacent vertices of $f_V(z)$. u and v are positive integers defined such that s_z has label (u) and s_{z+1} has label (v) .

Lemma 1.5. Rays $\overrightarrow{(u)(u+1)}$ and $\overrightarrow{(v)(v-1)}$ intersect or are parallel to each other.

Proof. Since $(u), (u+1), (v-1), (v)$ are in convex formation, then either $\overrightarrow{(u)(u+1)}$ and $\overrightarrow{(v)(v-1)}$ intersect or are parallel to each other, or else $\overrightarrow{(u+1)(u)}$ and $\overrightarrow{(v-1)(v)}$ intersect. If we suppose that the lemma statement does not hold for the sake of contradiction, then it follows that rays $\overrightarrow{(u+1)(u)}$ and $\overrightarrow{(v-1)(v)}$ must intersect. Let Q be the collection of vertices between v and u in W , going clockwise, inclusive. Since Q is in convex formation (it's a subset of W), and $\overrightarrow{(u+1)(u)}$ and $\overrightarrow{(v-1)(v)}$ intersect, it follows that $u+1$ is farther than u from any line joining two vertices of Q . Therefore removing s_z (which is the same vertex as (u)) from $f_V(z)$ and replacing it with vertex $(u+1)$ increases the area of $f_V(z)$, contradicting the assertion that $f_V(z)$ was the polygon of z vertices with maximal area on the set V . Therefore Lemma 5 is proven. $\square$

Lemma 5 allows us to apply Lemma 4 to the polygons formed by taking $(\mathcal{P}1234 \dots (2a))$ and subtracting out $f_V(a+1)$, as well as the polygons formed by taking $(\mathcal{P}1234 \dots (2a))$ and subtracting out $f_V(a-1)$. Adding the results tells us that

$$\begin{aligned} & 2[\mathcal{P}1234 \dots (2a)] - [f_V(a+1)] - [f_V(a-1)] \\ & \geq 2[\mathcal{P}1234 \dots (2a)] - [\mathcal{P}1357 \dots (2a-1)] - [\mathcal{P}2468 \dots (2a)], \end{aligned}$$

which is sufficient to prove Lemma 3. $\square$

Since $(\mathcal{P}135 \dots (2a-1)), (\mathcal{P}246 \dots (2a)) \leq f_V(a)$ since $(1), (2), \dots, (2a) \in V$. Using this fact with Lemma 3, we have proven Lemma 1 as desired. $\square$

Lemma 1.6. Any function g from integers to real numbers that satisfies $g(x+1) + g(x-1) \leq 2g(x)$ satisfies the inequality $g(x) + g(y) \leq g(x+1) + g(y-1)$ for $y-1 \geq x+1$, and is thus a convex function.

Proof. We proceed by induction on $y-x$. When $y-x=2$, the problem is trivial.

For $y-x > 2$, note that $g(x) + g(x+2) + g(y) + g(y-2) \leq 2g(x+1) + 2g(y-1)$, but the inductive statement shows that $g(x+2) + g(y-2) \geq g(x+1) + g(y-1)$. It follows that $g(x) + g(y) \leq g(x+1) + g(y-1)$, which completes our induction as desired. $\square$

It follows from the preceding lemma that $f_V(a)$ is a convex function, as desired.

2 Folding a paper

Let the smallest diameter be M . We claim that M is $\frac{\sqrt{2}}{2^{1006}}$. Let the P_0 be a 1 by 2 rectangle. For $0 \leq i \leq 2012$, polygon P_{i+1} is polygon P_i folded across line l_i . Line l_i splits P_i into P'_i and P''_i , and P''_i is reflected to P'''_i , and $P_{i+1} = P'_i \cup P'''_i$.

Claim 1. For $1 \leq i \leq 2013$, P_i can be expressed as the union of no more than 2^i convex polygons, and the average number of sides of these polygons is not more than 4.

Proof. We proceed by induction. P_0 clearly satisfies the inductive hypothesis. Suppose that P_i is the union of convex polygons $A_1, A_2, \dots, A_n$. Folding $A_1, A_2, \dots, A_n$ across l_i and taking the union of the resulting polygons yields P_{i+1} .

Let $S(P)$ be the number of sides of polygon P .

Lemma. A convex polygon Z is folded once. Z' and Z''' are convex, and $S(Z') + S(Z''') \leq S(Z) + 4$.

Proof. Any line l through Z splits Z into two convex polygons, and four new vertices are formed by l . The number of vertices in a convex polygon equals the number of sides, proving our lemma. $\square$

Thus P_{i+1} can be expressed as the union of no more than 2^{i+1} convex polygons, and the average number of sides of these convex polygons does not exceed 4. Claim 1 is proven. $\square$

Claim 2. For a quadrilateral of area A , the diameter has length less than $\sqrt{2A}$.

Proof. Without loss of generality, let the fixed area be 1. The diagonal is not longer than the diameter of the quadrilateral. Half of the product of the diagonals is not less than the area, and Claim 2 follows. $\square$

Claim 1 tells us that P_{2013} is the union of no more than 2^{2013} convex polygons, whose average number of sides is no more than 4. Part (a) of the problem tells us that there is some quadrilateral with area $\geq \frac{1}{2^{2012}}$, with vertices at the vertices of the final folded figure. Claim 2 tells us that the diameter of this quadrilateral, and thus of P_{2013} , is $\geq \frac{\sqrt{2}}{2^{1006}}$.